

Low-carbohydrate diet scores and risk of type 2 diabetes in men

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Background: Fat and protein sources may influence whether low-carbohydrate diets are associated with type 2 diabetes (T2D). Objective: The objective was to compare the associations of 3 low-carbohydrate diet scores with incident T2D. Design: A prospective cohort study was conducted in participants from the Health Professionals Follow-Up Study who were free of T2D, cardiovascular disease, or cancer at baseline (n = 40,475) for up to 20 y. Cumulative averages of 3 low-carbohydrate diet scores (high total protein and fat, high animal protein and fat, and high vegetable protein and fat) were calculated every 4 y from food-frequency questionnaires and were associated with incident T2D by using Cox models. Results: We documented 2689 cases of T2D during follow-up. After adjustments for age, smoking, physical activity, coffee intake, alcohol intake, family history of T2D, total energy intake, and body mass index, the score for high animal protein and fat was associated with an increased risk of T2D [top compared with bottom quintile; hazard ratio (HR): 1.37; 95% CI: 1.20, 1.58; P for trend < 0.01]. Adjustment for red and processed meat attenuated this association (HR: 1.11; 95% CI: 0.95, 1.30; P for trend = 0.20). A high score for vegetable protein and fat was not significantly associated with the risk of T2D overall but was inversely associated with T2D in men aged <65 y (HR: 0.78; 95% CI: 0.66, 0.92; P for trend = 0.01, P for interaction = 0.01). Conclusions: A score representing a low-carbohydrate diet high in animal protein and fat was positively associated with the risk of T2D in men. Low-carbohydrate diets should obtain protein and fat from foods other than red and processed meat.